Seminar2

luni, 18 octombrie 2021 12:03

AFD = un model de calcul

$M = (K, \Sigma, \delta, s, F)$

- K — mulțimea de stări
- Σ — alfabet
- δ — funcția de tranziție
- s — starea inițială
- F — mulțimea de stări finale

$0 < |K| < \infty$

$F \subseteq K$

$s \in K$

$|K| = 2.$

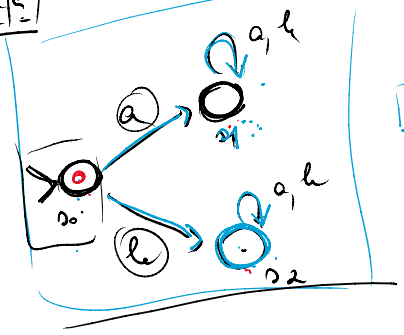
$|\Sigma| = 3.$

- un cuvânt este acceptat de AFD $\Leftrightarrow$ am parcurs toate simbolurile cuvântului și am ajuns într-o stare finală

$(s, w) \vdash^* (s', \varepsilon)$, $s' \in F$.

$w = a\,b\,b\,a$

$\Sigma = \{a, b\}$



$w = a\,b\,b\,a\,\varepsilon$ ✓

$w_2 \neq$ ✗

○ stare

>○ stare inițială

◎ stare finală

$p \xrightarrow{a} q$, $p, q \in K$, $a \in \Sigma$ $\quad \delta(p, a) = q$

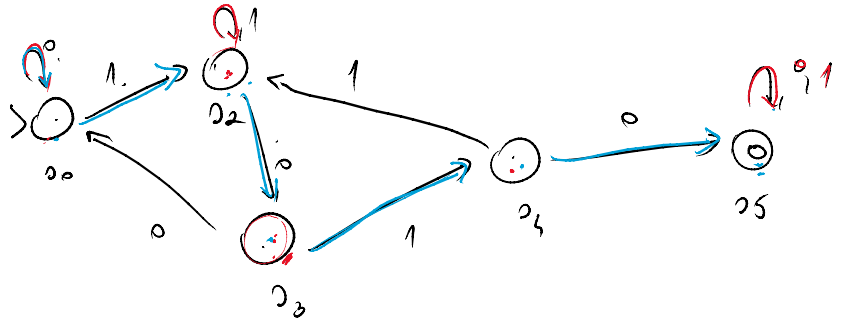


$\Sigma = \{0, 1\}$



$\Sigma = \{0, 1\}$

1.



$w_1 = 0\,1\,1\,0\,1\,0\,1$ ✓

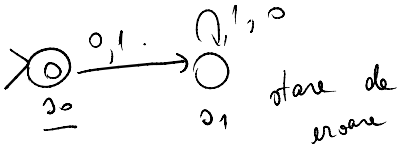
$w_2 = 0\,1\,0\,1\,1\,0$ ✗

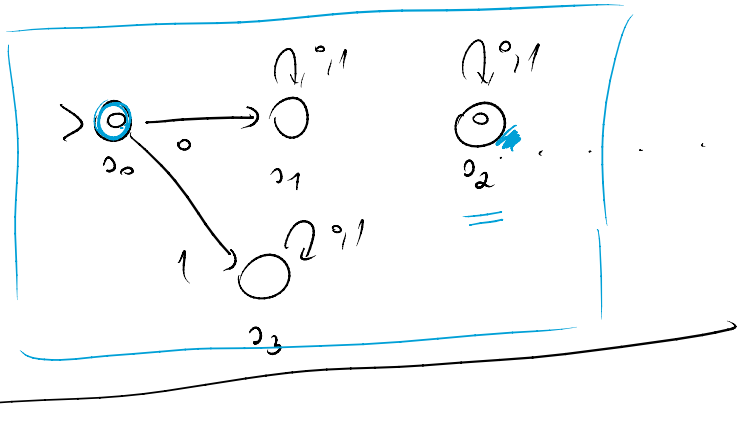
$(1 \cup 0)^* 1010 (1 \cup 0)^*$

2. $\Sigma = \{0, 1\}$

$L = \{e\}$

$w = e$

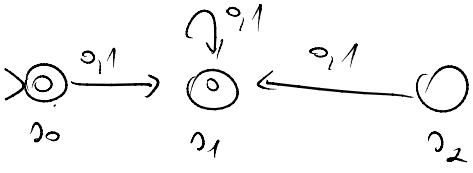






$\Sigma = \{0, 1\}$

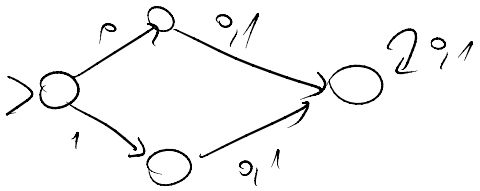
$L = \Sigma^*$

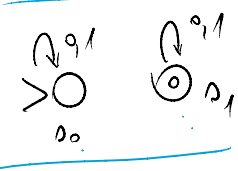


3. $\Sigma = \{0, 1\}$

$L = \emptyset$

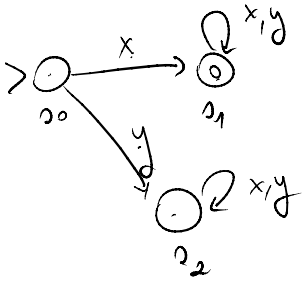






4. $\Sigma = \{x, y\}$

$L = \{ w \in \Sigma^* / \ w \text{ inicia en } x \}$



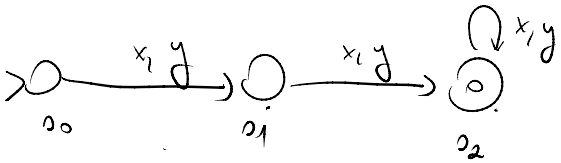
5. $\Sigma = \{x, y\}$

$L = \{ w \in \Sigma^* / \ \#(w) \geq 2 \}$

$w = abb$

$\#(w) = 3$.

$\#_b(w) = 2$.

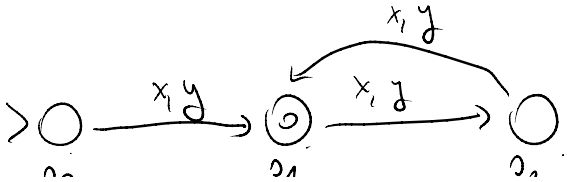


$w = xyy$ ✓

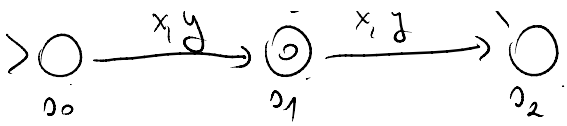
$w = yx$ ✗

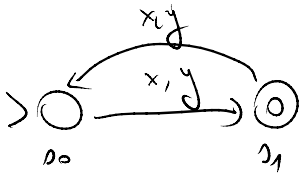
6. $\Sigma = \{x, y\}$

$L = \{ w \in \Sigma^* / \ \#(w) \not\geq 2 \}$



$w = xy$

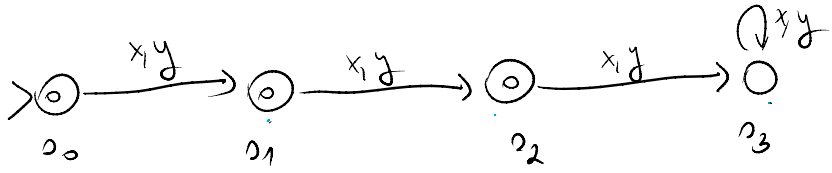




7. $\Sigma = \{x, y\}$

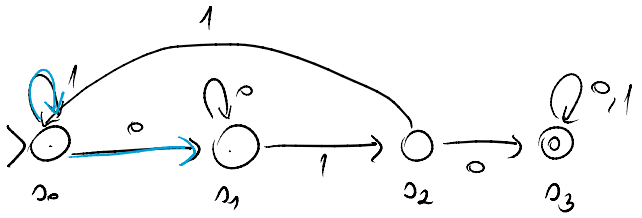
$L = \{w \in \Sigma^* / \#(w) < 3\}$

e,
x,
y,
xx,
xy,
yx,
yy



8. $\Sigma = \{0, 1\}$

$L = \{w \in \Sigma^* / \#_{010}(w) > 0\}$

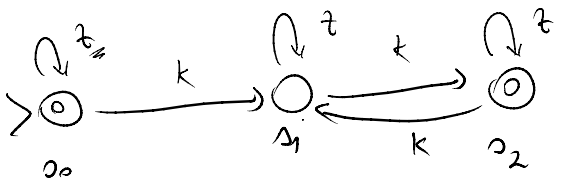


9. $\Sigma = \{k, z\}$

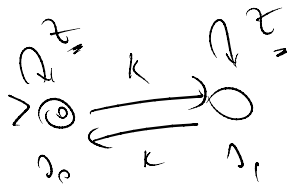
$L = \{w \in \Sigma^* / \#_k(w) : 2$ y w termina en $z\}$

10. $L = \{w \in \Sigma^* / \#_k(w) : 2\}$

$\cap z$ $\cap z$ + $\cap z$

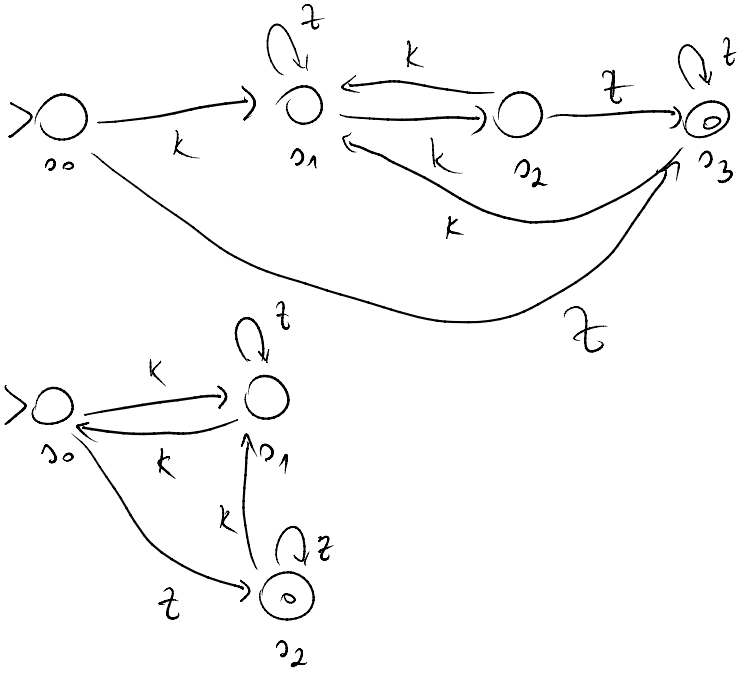


zz



9. $\Sigma = \{k, z\}$

$L = \{ w \in \Sigma^* / \#_k(w) \vdots 2$ y w termina en $z \}$



$\Sigma = \{0, 1\}$

$L = \{ w \in \Sigma^* / w$ no contiene $01 \}$